

SlouchFix

On-Device Posture and Screen-Distance Detection
for Programmers

The Programmer Posture Problem

- Several uninterrupted hours in front of a screen
- Deep focus makes posture deterioration easy to ignore
- Leaning toward the monitor increases neck strain and eye fatigue
- Poor desk habits accumulate gradually during long sessions



Why Current Solutions Fail

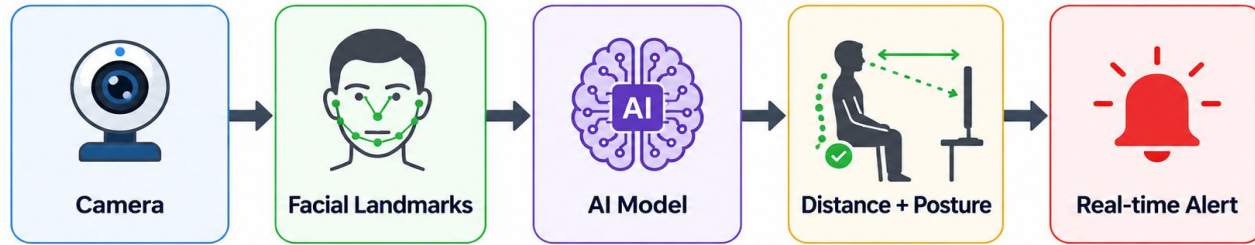
- Generic break reminders do not detect actual posture
- Wearables and dedicated sensors create additional cost and friction
- Rule-based camera tools may fail across users and desk setups
- May rely on external processing, raising privacy concerns

Can programmers improve posture without breaking focus
or sacrificing privacy?

Our Solution

SlouchFix

A lightweight desktop companion that monitors posture and viewing distance during coding sessions.



Features

Runs completely on-device

No raw video upload

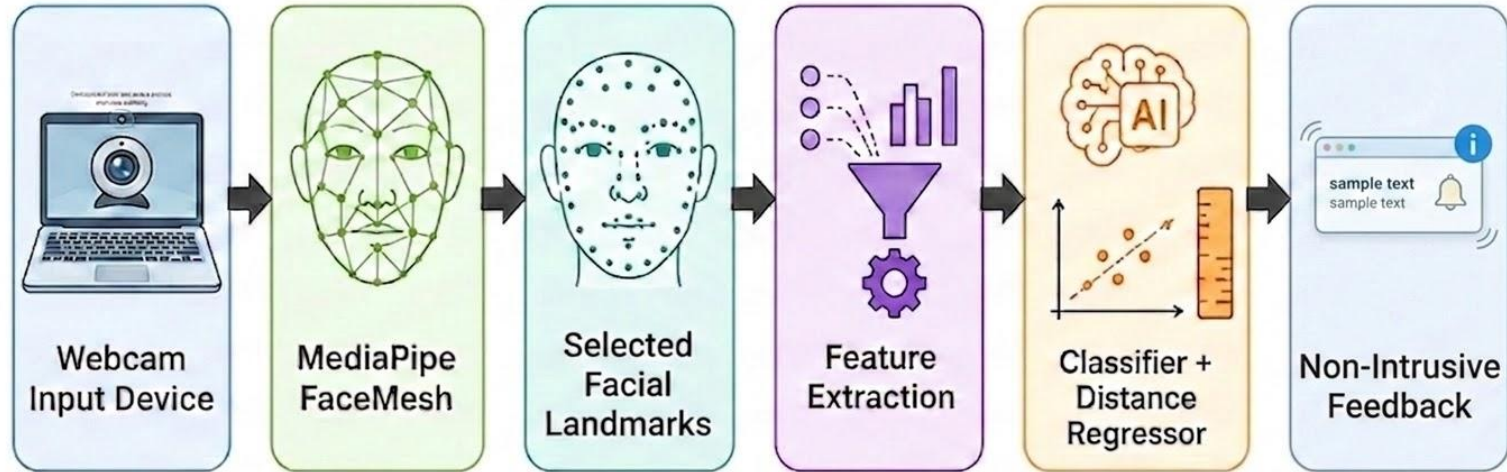
Real-time feedback

Lightweight processing

Non-intrusive desktop notifications

SlouchFix privately monitors posture and viewing distance during coding sessions using only lightweight facial-landmark features.

Methodology



Extracted Features:

- Head Pitch, Yaw, Roll
- Inter-Eye Distance
- Nose-to-Eye Ratios
- Face Bounding-Box Size
- Landmark Movement Over Time

Dataset

5-10 frequent computer users
Multiple monitor setups
Different lighting conditions
Multiple viewing distances
Different camera heights and angles

Labels

Good posture
Slouched
Leaning forward
Too close
Head tilted
Looking away

Model

Possible choices:

Rule-based threshold baseline

Random Forest

MLP

XGBoost

Output

Posture state

Estimated screen distance

Confidence score

Duration of poor posture

Evaluation

Metrics:

- Accuracy
- F1 Score
- MAE (cm)
- FPS
- Latency
- False alerts

Train



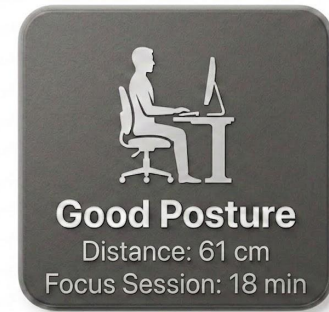
Validation



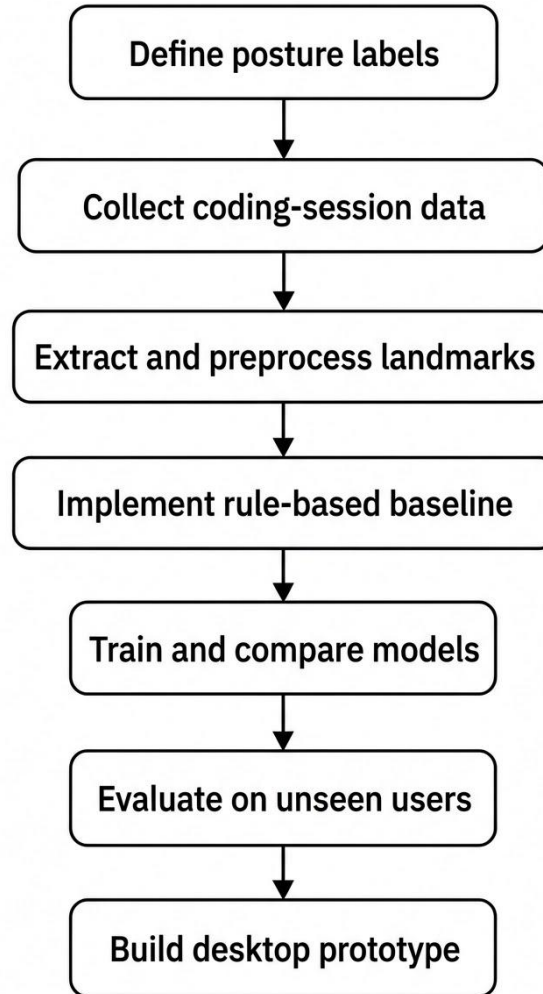
Test

Person-disjoint split

Expected Output



Timeline



Thank You!

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